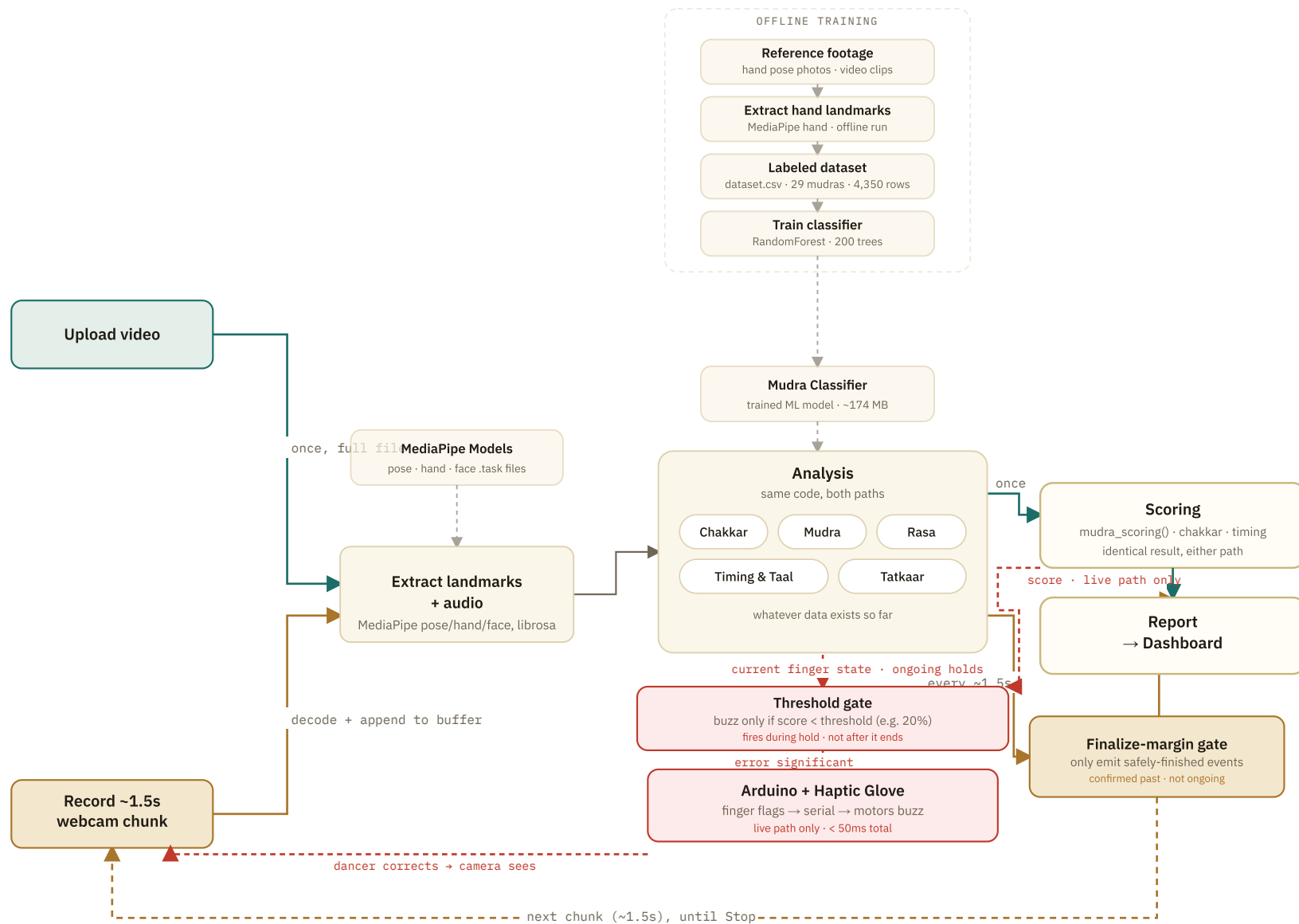


How a performance becomes a report

An uploaded video and a live webcam session enter differently and finish differently — but everything in between, the actual chakkar/mudra/rasa/timing/tatkaar analysis, is the exact same code running on the exact same shape of data.



Both entry points feed the identical analysis step. What differs is timing and framing: upload runs it once on a finished file; live re-runs it on a growing buffer roughly every 1.5 seconds, holding back anything still in progress (a spin or a mudra hold that hasn't finished yet) until it

clears a small safety margin, so the dashboard never shows an event as "done" and then unfinished a moment later.

Upload

One finished video, analyzed once, top to bottom. Simple, but you only see results after the whole file is processed.

Live

The browser records in ~1.5s chunks; each one is decoded, appended to a rolling buffer, and the full buffer is re-analyzed — so feedback updates as you dance, until you click Stop.

Hardware feedback

After scoring, per-finger mudra accuracy is read by a threshold gate — the same numbers shown on the dashboard. Only fingers below the threshold (e.g. 20% accuracy) trigger a buzz, so the glove and dashboard always agree. This prevents firing on minor deviations and lets the dancer converge. Once corrected, the camera sees the new hand position in the next chunk and the motor stops. A closed physical loop.

WHAT THE FIVE CHECKS ARE

Chakkar

Counts spins from shoulder rotation and scores landing quality — count accuracy, ending orientation, controlled vs. abrupt stop.

Tatkaar unvalidated

Detects rhythmic footwork as a sustained run of closely-spaced audio strikes — no camera view of the feet required.

Mudra

Identifies held hand gestures with a trained classifier, then checks the shape against reference rules.

Rasa unvalidated

Reads facial expression against the navarasa (nine classical emotions) on a fixed 1-second clock.

Timing & Taal

Checks wrist-movement peaks against the music's beat grid, and chakkar landings against the taal cycle's sam.

The live path, actually happening

Everything above is a diagram of what's *supposed* to happen in live mode. This is it actually happening — captured from a real live session, dancing in front of a webcam through this same dashboard.

• real live session

The finalize-margin gate, caught in the act

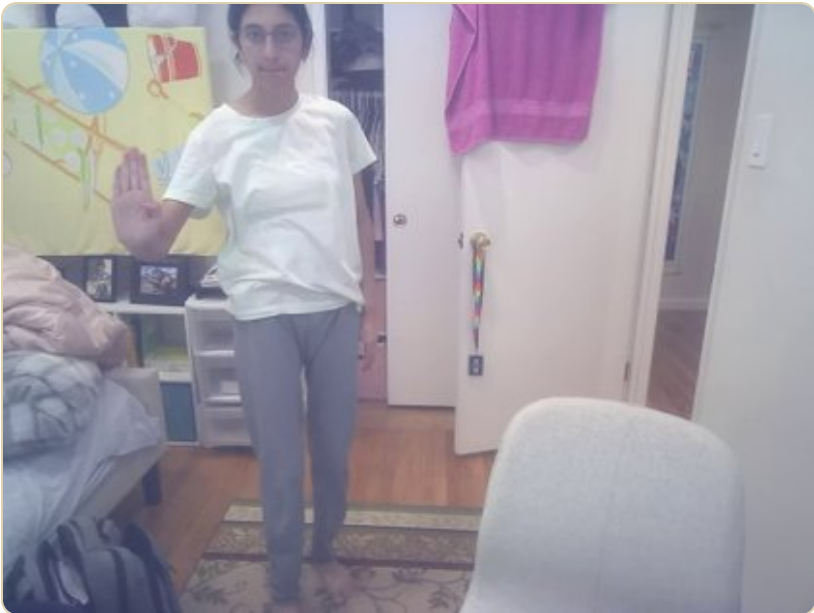
Three consecutive real ticks from one session. A second chakkar finishes mid-tick, sits **pending** for one cycle because it's still too close to the buffer's edge to trust, then clears the margin and gets **confirmed** — exactly the "don't report it, then un-report it" problem the gate exists to prevent.

5.95s	1st chakkar finalized — 78.4% quality	confirmed
7.44s	2nd chakkar detected (ended at 7.37s) — too close to report yet	pending
8.97s	same chakkar, now safely behind the margin — 100% quality	confirmed

real tick log · session 4f1534715bc1 · ~1.5s per tick

Real detections, live

The actual frame from a real live session, at the moment a held gesture was still sitting pending — and the mudras this same live pipeline has identified while recording, no upload, no waiting for the video to finish.



real frame · t=5.93s · the pataka hold below, mid-detection

pataka · left hand

4.55s – 5.95s

mushti · right hand

0.15s – 0.75s

tripataka · left hand

8.75s – 9.35s

real classifier output, two separate live sessions · identified
mid-session, not after Stop

Under the hood: real output, not mockups

Everything below was pulled from actually running this pipeline against a real uploaded performance (PTADanceVideo.mp4) and two of the recorded pilot clips — not staged or fabricated for this page.

• real pipeline output

Numeric precision

One tracked frame, 4.4 seconds into the real performance — every joint is a normalized (x, y) coordinate, not a category or a guess.

joint	x	y
right wrist	0.4886	0.2694
left wrist	0.5452	0.3559
right shoulder	0.4909	0.3407
right hand, wrist pt.	0.5737	0.3867
right hand, thumb tip	0.5396	0.3471
right hand, index tip	0.5238	0.3523

frame 132 · t=4.399s · from PTADanceVideo.csv, 3,417 rows total

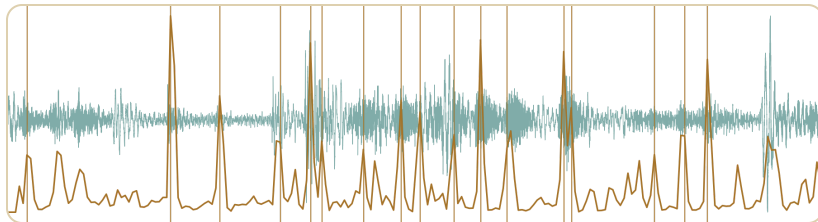
Skeletal tracking

The same moment, rendered — 33 body points plus 21 points per visible hand, tracked frame by frame.



Two signal types: image vs. audio

Mudra and pose come from pixels (above). Tatkaar comes from sound — a 5-second slice of the real performance's actual audio track, teal, with the detected onset-strength envelope and each flagged strike marked in gold.



raw audio waveform

onset strength + detected strikes

17 strikes detected in this 5-second window alone — part of a stretch the pipeline flagged at 280 strikes over 68 seconds.

The comparison algorithm, on real audio

No separate teacher/student pair exists in this project yet, so this runs the real comparison on two 34-second segments of the *same* performance — genuine algorithm output, not a real expert-vs-amateur pairing.

149

strikes, segment A

143

strikes, segment B

ratio **0.96** — flagged as not quite matching, the kind of near-miss a real teacher/student comparison is built to catch.

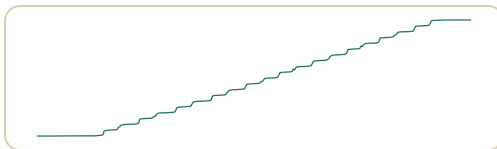
Inside Analysis: how each check actually works

The flowchart's "Analysis" box treats all five checks as one step. They're not one algorithm — each reads a different signal and asks a different question.

Chakkar

BODY POSE

Shoulder angle, tracked frame by frame, gets unwrapped into a continuous rotation signal. A sustained turn is segmented out as one spin event, and nearby segments within a small gap are merged so one spin doesn't fracture into several. Each event is scored on three things: **how close the rotation lands to a clean number of turns, how far the ending orientation drifts** from the starting one, and whether the stop was **controlled or abrupt** (from angular deceleration right at the end).

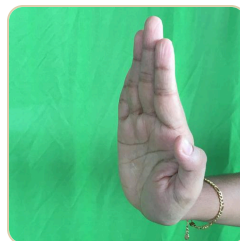


real signal · chakkar_03.mov ·
~10 turns over 12.1s

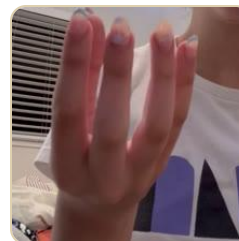
Mudra

HAND SKELETON

A mudra is a **held** pose, so it's found by looking for stillness, not motion: a window where hand-landmark movement drops and stays low long enough to count. Inside that window, finger-curl angles, thumb-to-fingertip distances, and finger-spread are averaged and handed to a small trained classifier, which names the closest-matching mudra. Separately, that mudra's own shape rule (which fingers should be extended vs. curled) is checked against the same measurements — so **rule agreement** (does the shape look internally consistent) and **identification accuracy** (was the guess actually right) are two different numbers.



trained on: Pataka
reference



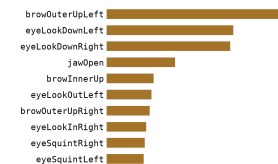
sees: a real held
gesture

Rasa

unvalidated

FACE MODEL

Sampled on a fixed one-second clock, not triggered by any particular movement. In each window, 52 named facial-expression parameters — from a separate face-tracking model, not the body/hand skeleton — are averaged and compared against reference criteria for each of the nine classical navarasas. A real smile paired with relaxed brows, for example, reads as **Hasya** (joy). The closest match is reported along with anything about it that didn't fit cleanly.

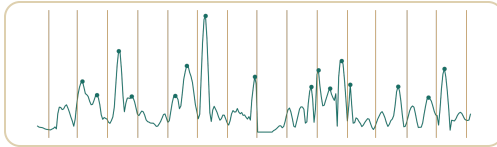


real blendshape scores · one
1s window · mudra_01.mov

Timing & Taal

BODY POSE + AUDIO

Both share one audio beat grid, detected once from the track's tempo. **Timing** checks whether gesture emphasis (peaks in wrist speed) statistically lines up with that grid over rolling 8-second windows, and separately flags a performance that ends noticeably off the music's last beat. **Taal** goes further, but only if you supply the taal and where the first sam falls: each chakkar's landing time is measured against the nearest sam using that taal's real cycle length (Teentaal's 16 beats, Jhaptaal's 10), not just the raw beat grid.



real wrist speed (dots = peaks) vs. real beat grid · 10s window

Tatkaar

unvalidated

AUDIO ONLY

Deliberately audio-only, not skeletal: a tatkaar's whole point is the percussive sound of the strike, and camera framing can't be trusted to keep feet in view anyway. The track's sound onsets are detected, then a **sustained run of closely-spaced onsets** — many strikes in a row at roughly the same pace — is reported as one phrase, with a strike count and rate, rather than trying to match individual bols one by one.



real detected stretches · 4
phrases across 114s ·
PTADanceVideo.mp4

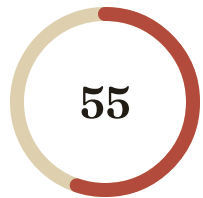
Score, report & dashboard: what a viewer actually sees

Real numbers from the same real performance — every flag above turns into a score here, and every score traces back to a specific, timestamped moment in the video.

Score aggregation

FLAGS → 0-100

Each check's flags collapse into a 0–100 score, and **Overall Score** averages whichever categories actually have data — a check with nothing to report (no mudra holds detected here) is left out, not counted as a zero.



Chakkar Quality

94.5%

Timing Accuracy

15.7%

Mudra

N/A

real scores · PTADanceVideo.mp4 · 8 chakkar, 113.9s

Report assembly

FLAGS → SENTENCES

Every flag from every check gets merged by timestamp into one chronological, plain-English list — the same list whether it came from chakkar, timing, or (once it's validated) tatkaar.

0:19 your chakkar was about 0.13 rotations short of a clean 1.0 (measured 0.87).

0:19 your chakkar did not return to your starting orientation — off by about 60 degrees to the left.

0:56 your movements were not clearly on beat from about 8s to 104s.

3 of 10 real flags · sorted by timestamp

Dashboard

THE DESTINATION

Both of the above land on the same screen — upload or live, the dashboard doesn't know or care which path the data took to get there.

Overview

Overall

55

Chakkar

94.5

Timing

15.7

Tatkaar

280

same real numbers, as they render on screen

